AROB Regression Report

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Introduction

In the context of practical works on the regression, we will compare three models representative of three approaches to non-linear function approximation: local approximation, basis function expansion and representation-learning approximation methods represented by Locally Weighted Regression (LWR), Radial Basis Function Network (RBFN) and Artificial Neural Network (ANN) respectively.

In this study, we measure mean squared errors (MSE) and training times of those models on datasets of varying size and noise level obtained from non-linear latent function. Figure 1 shows an example of such function and how these models fit it.

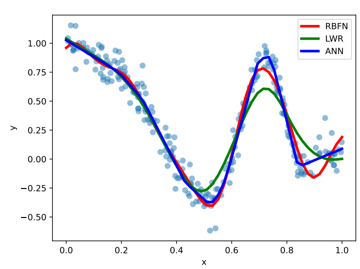

Figure 1: Fit of RBFN, LWR and ANN on example data, consisting of 250 points and generated with gaussian noise with standard deviation $\sigma = 0.1$.

Methods

Data

We generated the data using non-linear latent function $y = c_0 - x - \sin(c_1 \pi x^3)\cos(c_2 \pi x^3)e^{-x^4}$ where $c_0 \sim \mathcal{U}_{[0,2]}$ and $c_1, c_2, c_3 \sim \mathcal{U}_{[0,4]}$. To sample batches, we used a gaussian noise with 10 different values for the standard deviation σ linearly spaced between 0 and 0.5, and 7 different batch sizes: 100, 250, 500, 100, 200, and 5000.

Model Implementation and Hyperparameters

Among various implementations of RBFN algorithm, we have chosen batch RBFN using pseudo-inverse since, as we have established during the practical work, it performs fastest and mitigates numerical instabilities due to matrix inversion. We chose 10 Gaussian features as the hyperparameter, since it was a good compromise between fitting the example data and generalizing. For LWR, we have also taken an implementation using pseudo-inverse with 10 features.

The ANN that we studied was a Multilayer Perceptron (MLP) with 4 fully-connected hidden layers, each being comprised of 10 neurons with ReLU as activation function, those parameters have shown good training performances and required only around 300 iterations to converge with learning rate 0.003 (however it did not always optimally converge).

We chose to keep the hyperparameters mentioned above the same for each (size, noise) pair as finding the perfect hyperparameters for each combination would have been too computationally expensive.

Performance Evaluation

To get performance estimations of the models that run quickly (RBFN and LWR), we ran for each (size, noise) pair the models 5 times and calculated an average of and standard deviation of the performances (mean error and training time). It is important note that this study focuses on the quality of data fitting and measures only training MSE. We cannot thus infer on the generalization performance of the models, as it would require additional evaluation on a validation set, or an assessement of residuals with respect to the latent function itself, which is known in this setting.

Results and Discussion

Error Minimization and Noise Sensitivity

In the Figure 2 we can see that training MSE grows with increasing noise level for all three models. In general since our models fit the data quite well they come close the the mean squared error that the latent function would have had on the artificially generated data, estimating thus the noise itself giving a quadratic curve σ^2, where σ is the noise. All three models exhibit similar training MSE meaning that they fit the data equally well, however ANN shows higher variance of MSE estimates which can be due to initialization and convergence issues.

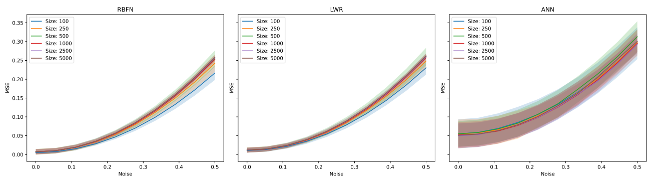

Figure 2: MSE as a function of noise level across different dataset sizes for three models. MSE increases with noise and smaller dataset sizes appear to be slightly less sensitive to noise for RBFN and LWR.

The dataset size doesn't have any apparent effect on MSE, except when the size is small (less than 250) where it has a small positive impact on performance of RBFN and LWR (Figure 3). It can be explained by the fact that smaller datasets are easier to overfit.

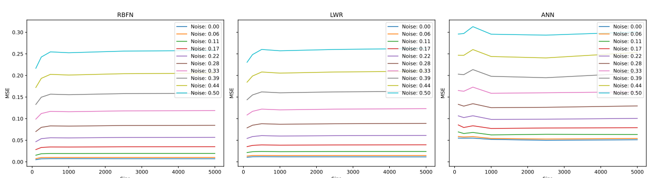

Figure 3: MSE as a function of dataset size across different noise levels for three models. MSE shows little to no dependence on dataset size across the explored range.

Computation Time Efficiency

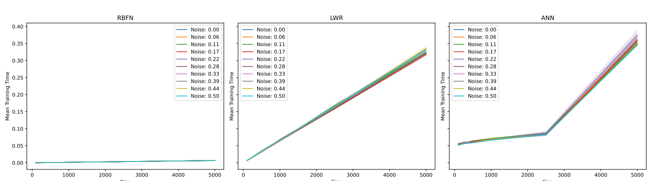

Figure 4: Training time as a function of dataset size across different noise levels for three models. Training time is larger for larger dataset sizes for LWR and ANN, but not for RBFN. Noise has negligible impact on the training times of all models.

As can be seen in Figure 4 the training time is affected by dataset size for all three models - as expected, larger dataset sizes require more time to be fitted. This relation is rather linear for LWR and non-linear for ANN, showing that ANN's training time is the most sensitive to dataset size. RBFN exhibits the fastest training among those three algorithms. It can be explained by the fact that RBFN uses only one expensive operation - pseudo inverse computation performing SVD, whereas LWR uses this same operation multiple times (once per feature), and ANN is an iterative algorithms requiring a lot of iterations and matrix multiplications.

Figure 4 shows that noise level in the data has a negligeable effect on training time. And, as can be seen in Figure 5, there is no apparent association between training time and MSE.

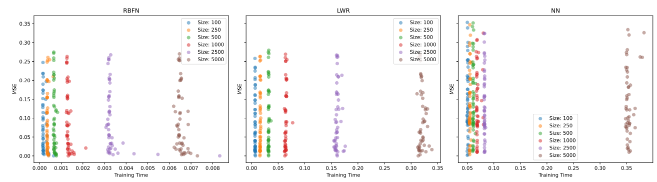

Figure 5: Distribution of MSE and training time across three models and different dataset sizes. There seems to be no strong association between MSE and training time.

Conclusion

All RBFN and LWR similarly well the non-linearity used in this study, ANN performing slightly worse. The quality of those approximations degrades when the training data is noisier, this trend being the same for the three models. Training time is mostly influenced by dataset size, and RBFN shows the best time performance.

We can thus conclude that for this type of non-linearity, RBFN seems to be the most efficient model. However, our results might be specific for the type of non-linear function we have generated (that can be simply approximated by a linear combination of relatively few Gaussians), and other models might perform better on more complex non-linearities. Another issue is that our study considered only single-dimensional input, so evaluating the performances of the models as the input dimension increases beyond 1 has not been considered.

Another limitation is that good training MSE can mean overfitting, thus a separate study on generalization of those models would be necessary.